

Youtube

Functional requirements:

user should be able to download/upload any videos based on his preference,

user should be able search for any video based on his preference,

Youtube app feed should get updated based on the user's activity

Non-functional requirements:

data-partitioning,

load balancing,

fault tolerance,

vertical scalability,

horizontal scalability,

low latency,

high throughput,

CAP theorem(Here eventual consistency is fine , but here we need 99.99% uptime

i.e Availability>>Consistency)

Core Entities:

user,

video,

channel,

feed

APIs:

POST:/upload

```
{
  userID,
  Filemetadata
}
```

GET:/search/{videoID}

HLD :

